

CS101

Binary + Java Variables

1/27

Agenda

1. cs50 Binary video
2. Counting in Binary (slides)
3. Alex Lee Tutorials on Variables
4. Quiz2-Practice (Kahoot)
5. Windows OS Tips + Touch Typing

Announcements

- Quiz1 Grades up
- Lab1 → Simplified, extended due date

Harvard cs50 Video on Binary/RGB

- <https://youtu.be/8mAITcNt710?si=h2bCwvG3kuXq55XE&t=314>
- Watch from **5:14** to **32:30**
- Google Search “Color Picker” to practice with RGB values

Binary Counting

Decimal Counting

100 10 1

X X X

Decimal Counting

10^2 10^1 10^0

X X X

Base 10

Dec → Ten (Latin)

Binary Counting

2^2	2^1	2^0
X	X	X

Base 2

Bi $\rightarrow$ Two (Latin)

Binary Counting

4	2	1
X	X	X

4-Digit Binary Counting

2^3 2^2 2^1 2^0

X X X X

4-Digit Binary Counting

8	4	2	1
X	X	X	X

4-Digit Binary Counting

8	4	2	1
0	1	0	0

4

4-Digit Binary Counting

8 4 2 1

1 0 1 0

10

4-Digit Binary Counting

8 4 2 1

1 1 1 1

15

4-Digit Binary Counting

0 0 1 1

3

4-Digit Binary Counting

0 1 0 1

5

Hexidecimal

16^2 16^1 16^0

X X X

- Base 16. Used by low-level programmers/security analysts
- Hexideci → Sixteen (Latin)
- Instead of 0-9, uses 0123456789ABCDEF

Hexidecimal

256 16 1

X X X

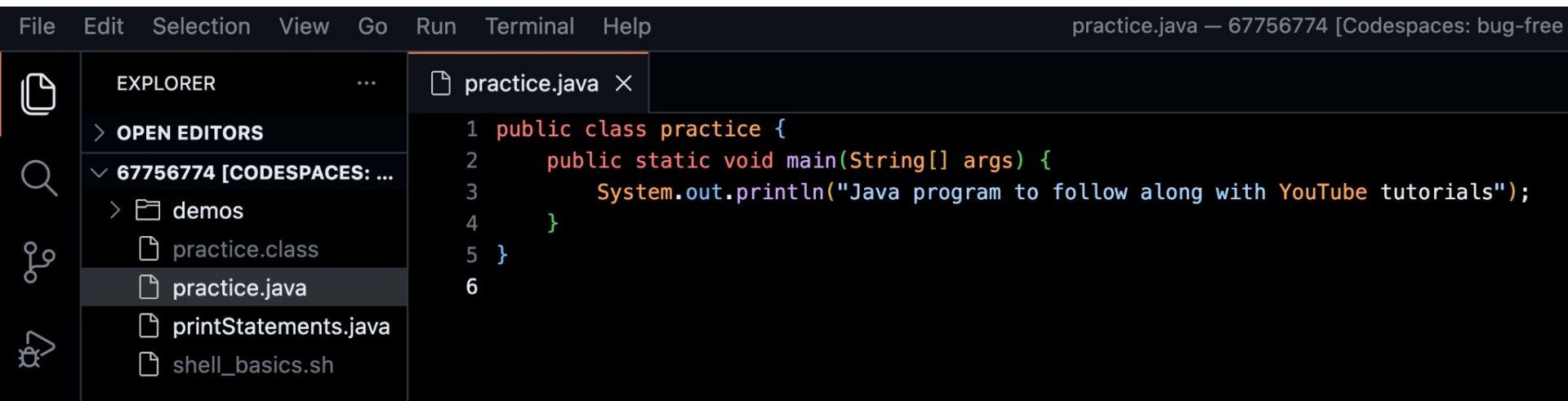
Hexidecimal

256	16	1
1	B	2

$$\begin{array}{r} 1 \times 2 \\ + 16 \times 11 \\ + 256 \times 1 \\ \hline = 434 \end{array}$$

Java Variables

Set up “practice.java” in Codespaces



Alex Lee Tutorials for Variables

- [#6 Java Strings Tutorial](#)
- [#7 Java Int Variables](#)
- [#9 Double, Long and Float Java Tutorial](#)

Set up lab3 in Codespaces

https://github.com/icanhazcodeplz/CS101/blob/main/labs/lab3_variables/lab3_variables_writeup.md

Quiz2 Practice

<https://github.com/icanhazcodeplz/CS101/blob/main/quizzes/Quiz2-Practice-Answers.pdf>

Next Class 1/29

Before Next Class:

- Finish Lab2 (Not technically due until Feb 12)
- Study for Quiz2
 - Alex Lee Videos #2, #3, #4
- Work through Alex Lee Videos #6, #7, #9

Next Class

1. Quiz2
2. Folder Structures
3. Java Arithmetic
4. Work on Lab3

- **Bring Headphones every class!**
- I'll bring food if everyone comes to an office hours or schedule a 1on1 **before Tuesday 2/3**